

Course Syllabus

ESCE-4560/6560: Modern Communication Systems

Fall 2025

Instructor:

Prof. Ish Jain

Electrical, Computer & Systems Engineering

Office: JEC 6018

Phone: 347-570-2502

Email: jaini@rpi.edu

Homepage: <https://faculty.rpi.edu/ish-jain>, <https://ishjain.github.io>

Office Hours: Tue/Fri after class or by Appointment

Teaching Assistant:

Shukai Chen

Electrical, Computer & Systems Engineering

Email: chens37@rpi.edu

Office Hours: TBD

Class Information

ECSE 4560 / 6560

CRN: 75325 / 75327

Time: Tue, Fri: 10:00-11:20am

Room: JEC 4104

Catalog Description:

A broad introduction to modern communication systems, with applications to emerging wireless communication technologies. Fundamental principles of wireless channel modeling; performance limits of communication systems; analog and digital modulation techniques; multi-antenna communication systems; multi-carrier modulation; multi-user communication systems. Emerging wireless technologies such as MIMO and OFDM, with applications to cellular and wireless LAN communication standards. Emerging wireless sensing applications such as localization, tracking, and FMCW Radar sensing. Understanding time, frequency, and space as fundamental dimension for communication and sensing. Complemented with computational exercises.

Prerequisites:

ECSE-2410 (Signals and Systems) and ECSE-2500 (Engineering Probability) or equivalent.

Text:

Required

Goldsmith, A. (2005). Wireless Communications, Cambridge University Press. (available as e-book through RPI library)

References

- Tse, D., and P. Viswanath (2005). Fundamentals of Wireless Communication, Cambridge University Press. (available as e-book through RPI library)
- Proakis, J., and M. Salehi (2008). Digital Communications (5th ed.). McGraw Hill.

- Molisch, A. F. (2011). Wireless Communications (2nd ed.). Wiley Publishing.

Course Content (Tentative):

The objective of the course is to provide an introduction to analog and digital communication techniques, with applications to modern wireless communication systems and technology standards. *Since this is an advanced topics course, the following schedule and the list of topics are tentative, and subject to change depending upon the progress of the class.*

Specific topics include:

Lecture #	Topic
1	Intro to wireless communication and sensing systems
2,3	Signals review, channel modeling, modulation schemes
4,5	Communication Architecture
6,7,8,9	Multi-carrier modulation and OFDM
10,11	Multi-antenna systems and MIMO, mmWave beamforming
12,13	Multi-user systems (TDMA, FDMA, CDMA, OFDMA)
14	Exam 1
15,16	Fundamentals of Sensing: Space Fourier Transform
17,18	Monostatic Radar FMCW sensing
19, 20,21	Bistatic Sensing – localization and tracking with WiFi
22,23,24	Cellular communication standards (3G, 4G/LTE, 5G, 6G) and WiFi
25	Exam 2
26,27,28	Other Emerging Topics and Final project presentations

Student Learning Outcomes:

Students of who successfully complete this course (either 4560 or 6560) will be able to:

- **Analyze** different analog and digital modulation schemes in terms of their performance advantages and disadvantages.
- **Describe** how OFDM and MIMO operate and are used in cellular communication standards.
- **Contrast** the relative advantages and disadvantages of different multi-access schemes for multi-user wireless systems.
- **Develop** understanding of emerging topics in wireless sensing
- **Understand** fundamentals of time, frequency, and space dimensions for communication and sensing systems

Students of who successfully complete 6560 will additionally be able to:

- **Model** communication systems using mathematical tools and **analyze** their performance under different channel properties.
- **Assess** the current limits of theory, knowledge and practice in analyzing emerging communication systems, and **propose** new research directions through a term paper.

Course Assessment Measures:

Assessment for ECSE 4560

- Attendance and In-class participation: 5% of the grade
- Per-class Quizzes: 5% of the grade
- *Homeworks and MATLAB Assignments:* 40% of the grade

A total of 8 assignments, assigned weekly, with regular homeworks and programming (MATLAB/LabVIEW) assignments. The homeworks will include some qualitative and quantitative questions, and the MATLAB/LabVIEW assignments would require some programing using these tools.

- *Midterm Exams:* 2 exams, 30% of the grade (20%, 10% each exam)

Midterm exams will be conducted during regular class hours. The midterm exams are not comprehensive; Midterm Exam 1 will cover channel modeling, modulation, OFDM, MIMO Midterm 2 will cover wireless sensing applications. Exams will comprise of a mix of qualitative and quantitative questions – similar to the homeworks but without the MATLAB and/or LabVIEW assignments.

- *Final Project:* 20% of the grade

Student-defined project proposal or choice from list provided by instructor at the beginning of the semester. Students can build upon the MATLAB or LabVIEW assignments assigned in the homeworks into a more comprehensive project. Can be done individually or in groups of two. Students will be expected to make a presentation on the project at the end of the semester, submit a report and perform peer-evaluation.

Assessment for ECSE 6560

- Attendance and In-class participation: 5% of the grade
- Per-class Quizzes: 5% of the grade
- *Homeworks and MATLAB/LabVIEW Assignments:* 40% of the grade

A total of 8 assignments, assigned weekly, alternating between regular homeworks and programming (MATLAB/LabVIEW) assignments. The homeworks will include some qualitative and quantitative questions, and the MATLAB/LabVIEW assignments would require some programming using these tools.

In the homeworks, ECSE 6560 students will be assigned one or more problems/subproblems that are more advanced than those assigned to ECSE 4650 students, requiring deeper thinking and focusing on applying mathematical tools to analyze performance of communication systems.

- *Midterm Exams:* 2 exams, 20% of the grade (10%, 10% each exam)

Midterm exams will be conducted during regular class hours. The midterm exams are not comprehensive; Midterm Exam 1 will cover channel modeling, modulation, OFDM, MIMO Midterm 2 will cover wireless sensing applications. Exams will comprise of a mix of qualitative and quantitative questions – similar to the homeworks but without the MATLAB and/or LabVIEW assignments.

- *Final Project (includes term paper):* 30% of the grade

Student-defined project proposal, in consultation with the instructor. *Students will assess the limitations of the current state-of-the-art for analyzing emerging communication systems and implement these techniques on a new system (e.g., Matlab toolbox) or dataset (e.g. NIST, ASU, Colosseum dataset).* Can be done individually or in groups of two. Students are expected to report their findings and observations as a *term paper* in addition to making a presentation towards the end of the semester.

Course Policies:

- **Quiz/Homework:** Per-class Quizzes will be assigned on LMS before every lecture (Lecture 2 onwards) and will constitute of simple questions that will require you to pay attention to the material discussed in class. They will be due the end of day (11:59pm on the same day). Homework/Lab assignments will be posted on Piazza. Students need to submit their solutions on Gradescope by the deadline. These assignments will be posted roughly on a weekly basis.
- **Integrity:** All assignments are to be completed on your own. You are allowed to consult with other students in the current class or the instructor (e.g. through Piazza) regarding the conceptualization of the problem and possible methods of solution, but you may not share details or your final answer with other students. You may not copy solutions from anyone or any source. Failure to follow the rules will be considered violation of academic integrity.
- **Lateness:** Late submission of assignments are generally not allowed, and late submission without prior instructor permission will not be graded. Exception-1: Submissions are typically due on Friday and a late submission until Monday will incur a 20% penalty. Exception-2: You are allowed a total of 2 late submissions of your choice until Monday free of penalty.
- **Class attendance/participation:** Class attendance is not mandatory, and class participation is graded. Students are highly encouraged to attend lectures and ask questions during and after class. The lectures will be video-recorded, and handwritten notes made during the lectures will be posted on Piazza after each class.

- **Missed exams or assignments:** An approved absence is one that has been approved following Rensselaer's rules and procedures, and should be documented, typically via a letter from the dean's office, and emailed beforehand to the instructor. A missed exam due to an unexpected emergency should be communicated in writing as soon as possible.
- **Use of Wireless Devices:** Use of cell phones during the lectures or exams is not allowed. Use of other computing devices such as laptops, iPads, etc. is generally discouraged during class hours. Exams are open book, open notes, but no direct or indirect communication with any person other than the instructor is allowed. Any illicit information exchange during an exam will be considered a violation of academic integrity and will be dealt accordingly.

Instructor access and Discussion Boards:

Outside of class hours, students can access the instructor in one of the following ways:

- **Regular office hours:** Right after class, from 11:30am-12:30pm Tue and Fri.
- **By appointment:** You can email the instructor for meeting (in-person or on WebEx) outside of the regular office hours.
- **Piazza:** Students can ask questions, as well as discuss homeworks and other course related issues with their classmates on Piazza. The TA and instructor will monitor the messaging board and respond as needed.

The TA will conduct office hours as well, the details of which will be announced. The TA will be grading the homeworks and the exams. If you have a grading issue, you can request re-grading on Piazza; or contact the instructor if it remains unresolved.

The instructor plans to seek feedback from the students during the semester to assess how the instruction is progressing, and if the students have any specific concerns regarding the course. Unsolicited feedback on any general concerns, and suggestions on how student learning can be improved, are also welcome.

Academic Integrity:

Student-teacher relationships are built on trust. For example, students must trust that teachers have made appropriate decisions about the structure and content of the courses they teach, and teachers must trust that the assignments that students turn in are their own. Acts that violate this trust undermine the educational process. The Rensselaer Handbook of Student Rights and Responsibilities and The Graduate Student Supplement define various forms of Academic Dishonesty and students must make themselves familiar with these. In this class, all assignments that are turned in for a grade must represent the student's own work. In cases where help was received, or teamwork was allowed (such as the final project), a notation on the assignment should indicate your collaboration.

In general, while discussions with your classmates are allowed (and encouraged!) for homeworks, the work submitted for grading must be entirely the student's work. A similar policy applies to any MATLAB and/or LabVIEW assignments as well. For exams, no information exchange with anyone (other than the instructor) is allowed. A grade of zero will be given on the first assignment where a violation is detected, and a referral to the appropriate Dean (Dean of Students or the Dean of Graduate Education for undergraduate or graduate students, respectively). If there is a subsequent infraction the student will receive a grade of "F" for the course and referral to the appropriate Dean.

Academic Accommodation:

Rensselaer Polytechnic Institute is committed to providing equal access to our educational programs and services for students with disabilities. If you anticipate or experience academic barriers due to a disability, please contact the Office of Disability Services for Students (DSS) (dss@rpi.edu; 518-276-8197) to establish reasonable accommodations. Once you have been approved for accommodations, please provide your Faculty Memorandum (a letter provided to students by DSS) to all faculty members for this course. Please provide this at the very beginning of the semester.